IST 691: Research Methodology and Scientific Writing Skills

Hypothesis Testing

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Outline

- Data Collection
- Statistics – Review of Some Fundamental Concepts
- Inferential Statistics
- Hypothesis Testing

Data Collection Issues

- Sample Size
- Procedure
- Institutional Approval
- Participant Consent
- Debriefing (Postexperimental Interview)

Data Collection Issues - Sample Size

- Sample size is very important for getting accurate, statistically significant results
- Small sample size
  - You may include a disproportionate number of individuals which are outliers and anomalies
  - These skew the results and you don’t get a fair picture of the whole population
- Sample size too large
  - Study becomes complex, expensive and time-consuming
  - Results are more accurate, but the benefits don’t outweigh the costs

Computing Sample Size – Known Population Size

- Need to determine the following variables
  - Population size (if known)
    - Population size refers to the total number of people in your target audience
  - The margin of error (confidence interval)
    - Refers to how closely your sample size accurately represents your target audience
    - Smaller margin of error means a higher level of accuracy
    - Larger margin of error means being less accurate. There will always be some margin of error, but you can control how big, or small, your sample size is
  - Confidence level
    - Related to the margin of error, confidence level refers to how accurate you feel your results are
    - Expressed as a percentage, states how sure you are that the mean falls somewhere within the margin of error
    - Most commonly used values: 90%, 95%, or 99%
  - Standard deviation
    - A measure of how dispersed or spread out data values are relative to their mean (average).
    - Low standard deviation means scores will all be clustered near the mean, so not much variation
    - Larger standard deviation means when plotted on a graph, responses will be more spread out
    - 0.5 is a good standard deviation to set to ensure a proper sample size

Computing Sample Size – Known Population Size

- Steps in computation
  - Step 1
    - Determine the population size, N
      - In case the population size is very large but the exact number is not known, use 100,000 because the sample size doesn't change much for populations larger than that.
  - Step 2
    - Determine the z score (explained later) of the normal distribution at the required confidence level
    - Common Z-scores
      - 90%: Z Score = 1.282; 95%: Z Score = 1.645; 99%: Z Score = 2.326 (1 tail)
      - 90%: Z Score = 1.645; 95%: Z Score = 1.96; 99%: Z Score = 2.576 (2-tails)
  - Step 3
    - Next, determine the sample proportion which can be used from previous similar studies or be collected by running a small pilot survey
      - If unsure, one can always use 0.5 as a conservative approach, and it will give the largest possible sample size
  - Step 4
    - Determine the margin of error, the range in which the true population expects to lie
  - Step 5
    - Use the formula for sample size (see next 2 slides)

Computing Sample Size – Known Population Size

$$n = N \times \frac{\frac{Z^2 \times p \times (1-p)}{e^2}}{\left[N - 1 + \frac{Z^2 \times p \times (1-p)}{e^2}\right]}$$

N = Population size.
Z = Z value
p = Sample proportion.
e = Margin of error

- Example
  - A website owner is interested to know how many of his customers bought an item from him after viewing his website on a certain day. Given that the website has on average 10,000 views per day, determine the sample size of the customers that he has to monitor at a 95% confidence level with a 2% margin of error.
    - N is 10,000, z value is 1.96, e = 0.02
    - Assume p = 0.5

$$= 10{,}000 \times \frac{\frac{1.96^2 \times 0.5 \times (1 - 0.5)}{0.02^2}}{10{,}000 - 1 + \frac{1.96^2 \times 0.5 \times (1 - 0.5)}{0.02^2}}$$

$$= 194$$

Computing Sample Size for Unknown Population Size

$$\text{Necessary sample size} = \frac{(\text{Z-score})^2 \times \text{StdDev} \times (1 - \text{StdDev})}{(\text{margin of error})^2}$$

- Example
  - For a confidence level of 90%, the recommended standard deviation of 0.5 and a margin of error of +/- 5%

$$\frac{(1.645)^2 \times 0.5 \times (1 - 0.5)}{(0.05)^2}$$

$$\frac{2.7 \times 0.5 \times 0.5}{0.0025} = \frac{0.675}{0.0025} = 270$$

Data Collection Issues - Procedure

- A pilot study may be essential
  - A pre-test (run-through) of the experiment with a small number of participants
  - Advantages
    - Poorly worded instructions will become apparent
    - May give an indication as to whether the independent variable manipulation produces the intended effect
    - Gives the researcher experience with the procedure

Data Collection Issues - Institutional Approval

- Some experiments, especially those involving animals and humans require institutional approval
  - Institutional Animal Care and Use Committee reviews studies to determine if animals are used in appropriate ways
    - Avoiding or minimizing pain and discomfort to animals
    - Whether activities involving surgery include appropriate pre- and post operative care
    - Etc.
  - Institutional Review Board reviews studies involving human subjects to determine if the procedures used are appropriate
    - Primary concern: welfare of human participants
      - Issues related to informed consent, ensuring that procedures used do not harm the participants

Data Collection Issues - Participant Consent

- Research participants must give their consent to participate in the study, except the study falls under the exempt category
  - Exempt category includes interviews, survey procedures, observation of public behaviours in which the participant in anonymous, and a number of others
- Consent is given in a *consent to participate* statement
  - Idea is to provide research participants with complete information about the study so that they can make an intelligent and informed choice as to whether they want to participate

Data Collection Issues
Debriefing (Post-experimental Interview)

- This is an interview with the participant following completion of the experiment during which all aspects of the experiment are explained and the participant is allowed to comment on the study

Outline

- Data Collection
- Statistics – Review of Some Fundamental Concepts
- Inferential Statistics
- Hypothesis Testing

Statistics – Review of Some Fundamental Concepts

- Central Tendency
- Variance and Standard Deviation
- The Normal Distribution & Standard Scores
- Inferential Statistics

Central Tendency

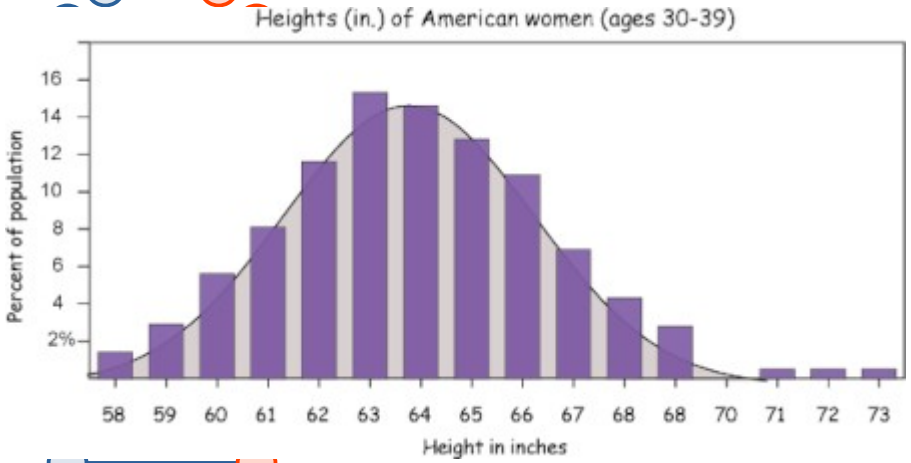
- Measures of central tendency are summary statistics that represent the centre point or typical value of a dataset
- Most common measures: mean, median, and mode
- Mean
  - The mean is the arithmetic average
- Median
  - The middle value
  - Splits the dataset in half
- Mode
  - The value that occurs the most frequently in your data set

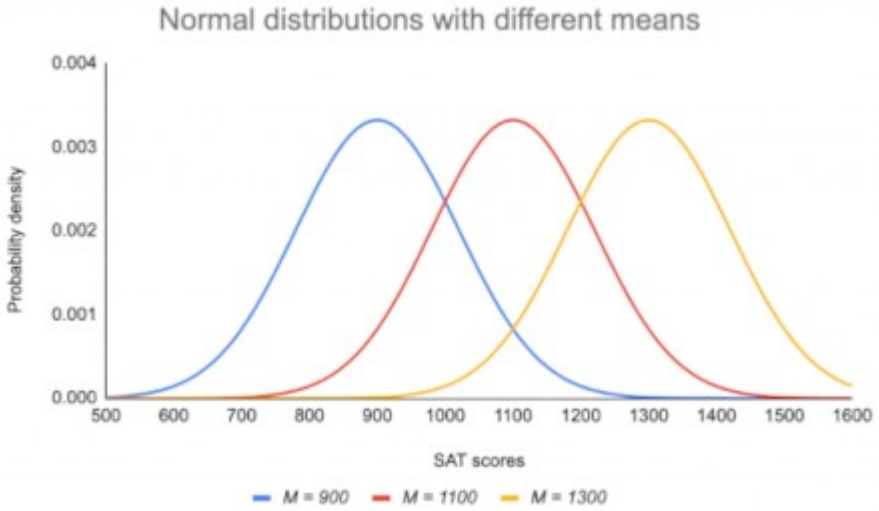
Variance and Standard Deviation

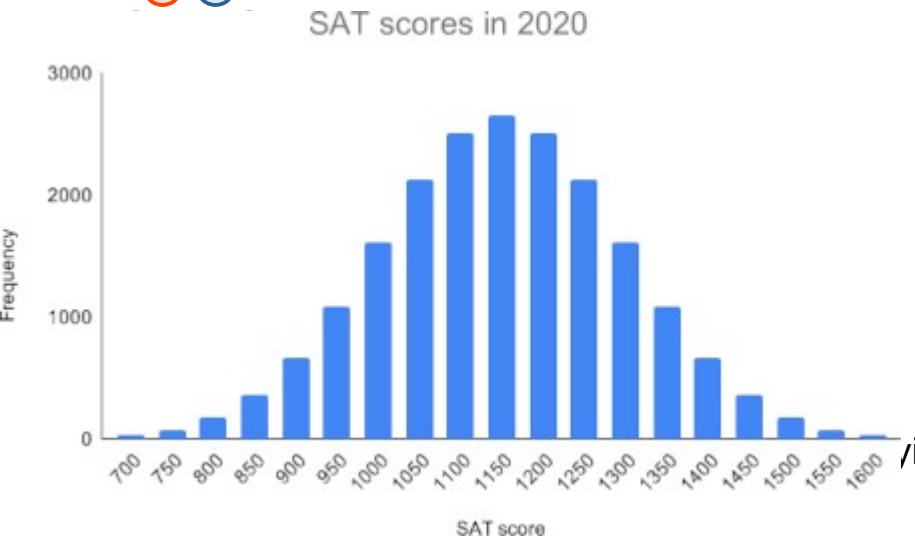
- Variance
  - A measure of how far a set of data are dispersed out from their mean or average value
    - Population variance is denoted as ‘σ^2’
    - Sample variance is denoted as ‘s^2’
- Standard deviation
  - A measure of the spread of statistical data
  - Measures the deviation of data from its mean or average position
    - Population SD denoted by the symbol, ‘σ’
    - Sample SD is denoted as ‘s’
- Variance vs standard deviation
  - Variance gives results in squared units, while standard deviation in original units
    - Standard deviation is usually more useful to describe the variability of the data
    - Variance is usually much more useful mathematically

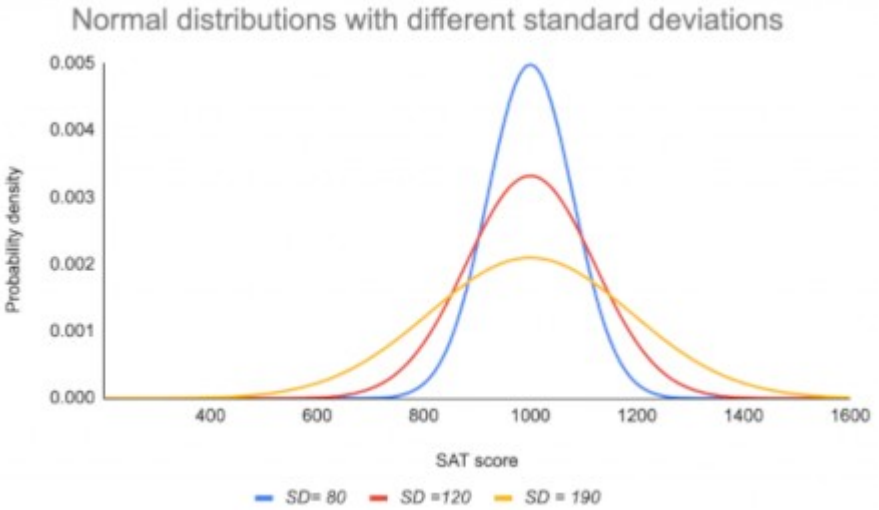
Importance of the Normal Distribution

- Also known as the Gaussian distribution, is the most important probability distribution in statistics for independent, random variables
- It is a continuous probability distribution that is symmetrical around its mean, most of the observations cluster around the central peak, and the probabilities for values further away from the mean taper off equally in both directions
  - Extreme values in both tails of the distribution are similarly unlikely
- As with any probability distribution, the normal distribution describes how the values of a variable are distributed
- Importance derives from the fact that it accurately describes the distribution of values for many natural phenomena
  - Heights, blood pressure, measurement error, IQ scores, etc.


Heights (in.) of American women (ages 30-39)
Percent of population
16
14
12
10
8
6
4
2%
58 59 60 61 62 63 64 65 66 67 68 68 70 71 72 73
Height in inches


Normal distributions with different means
Probability density
0.004
0.003
0.002
0.001
0.000
500 600 700 800 900 1000 1100 1200 1300 1400 1500 1600
SAT scores
M = 900
M = 1100
M = 1300

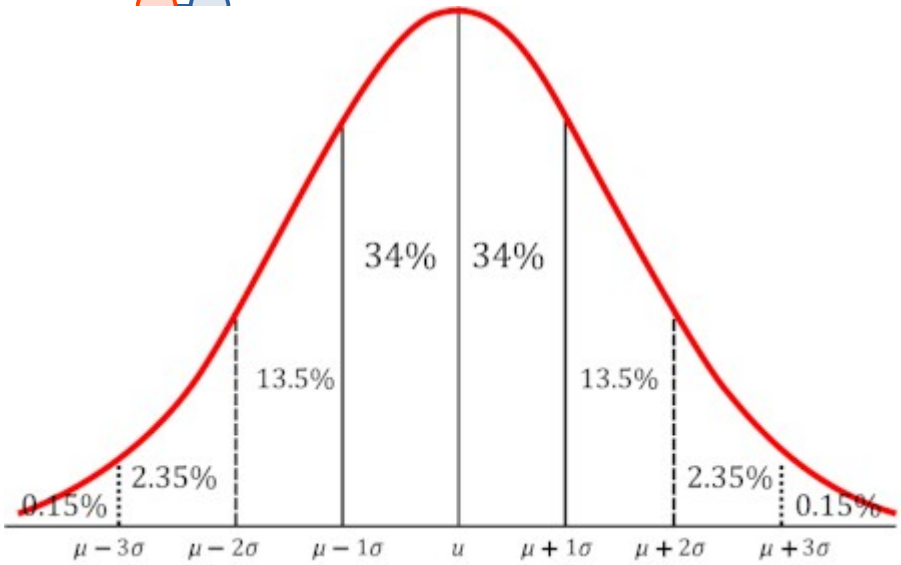

SAT scores in 2020
Frequency
3000
2000
1000
0
700 750 800 850 900 950 1000 1050 1100 1150 1200 1250 1300 1350 1400 1450 1500 1550 1600
SAT score


Normal distributions with different standard deviations
Probability density
0.005
0.004
0.003
0.002
0.001
0.000
400 600 800 1000 1200 1400 1600
SAT score
SD= 80
SD =120
SD = 190

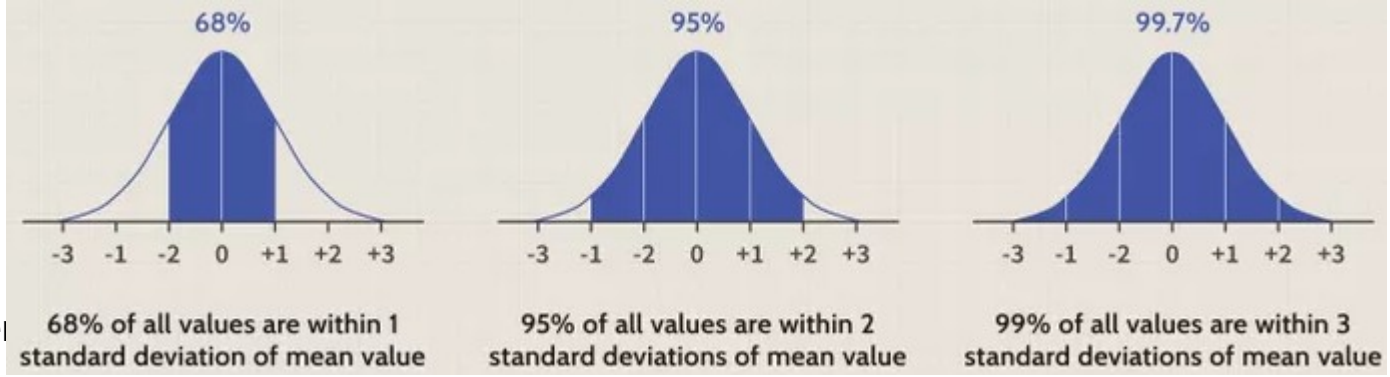
Properties of Normal Distribution

- Symmetric bell curve
  - The distribution falls symmetrically around the mean, the width of which is defined by the standard deviation
- Its mean (average), median (midpoint), and mode (most frequent observation) are all equal to one another
  - These values all represent the peak, or highest point, of the distribution
- Half of the population is less than the mean and half is greater than the mean
- The Empirical Rule applies (see next slide)

Empirical Rule

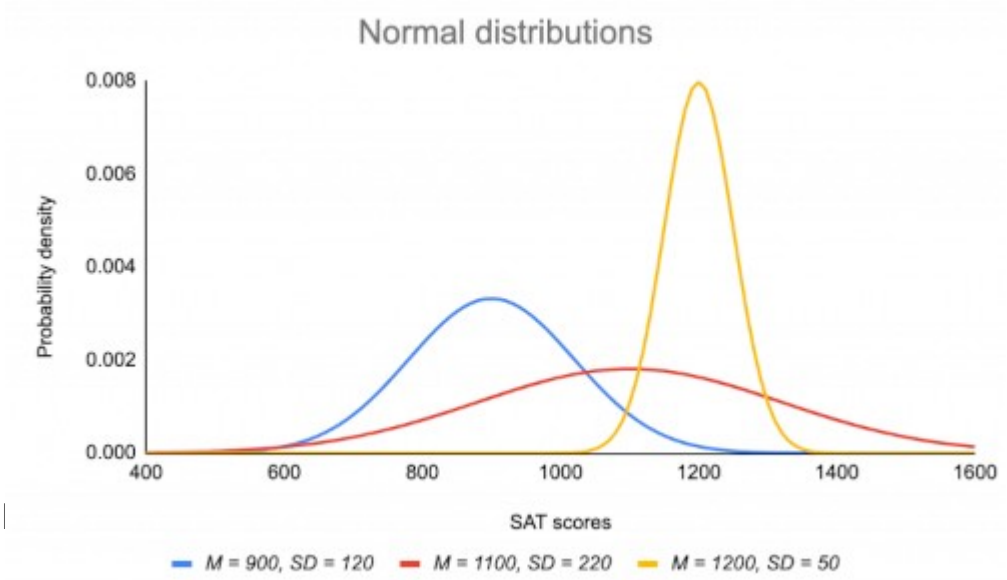


- For all normal distributions
  - 68.2% of the observations will appear within plus or minus one standard deviation of the mean
  - 95.4% of the observations will fall within +/- two standard deviations
  - 99.7% within +/- three standard deviations
- Can be seen that data falling outside of three standard deviations ("3-sigma") would signify rare occurrences



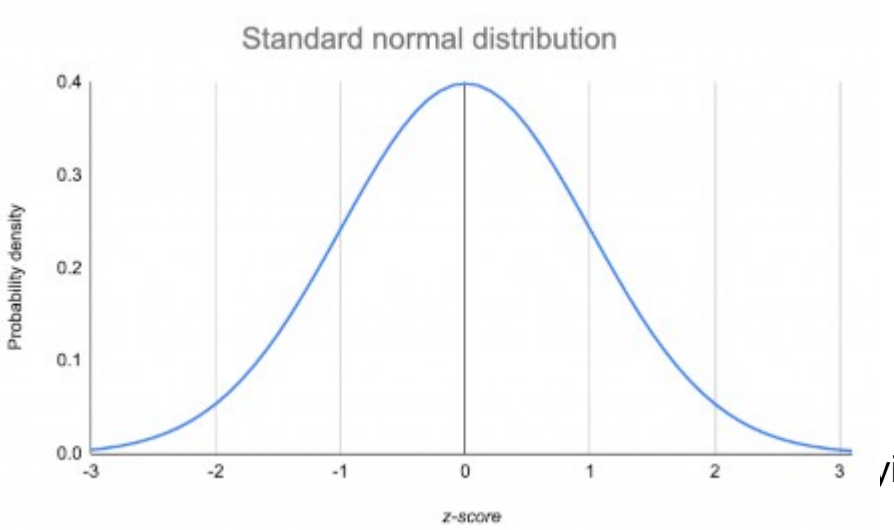
The Standard Normal Curve

- Also called the z-distribution, is a special normal distribution where the mean is 0 and the standard deviation is 1
- Every normal distribution is a version of the standard normal distribution that's been stretched or squeezed and moved horizontally right or left
  - A small standard deviation results in a narrow curve, while a large standard deviation leads to a wide curve



The Standard Normal Curve

- While individual observations from normal distributions are referred to as x, they are referred to as z in the z-distribution
- Every normal distribution can be converted to the standard normal distribution by turning the individual values into z-scores.
- Z-scores tell you how many standard deviations away from the mean each value lies



Z-score Formula	Explanation
$z = \frac{x - \mu}{\sigma}$	• x = individual value • μ = mean • σ = standard deviation

About Z-Score

- Z score
  - Also known as the standard score, indicates how many standard deviations an entity is, from the mean
  - A way to compare the results from a test to a “normal” population
  - If X is a random variable from a normal distribution with mean (μ) and standard deviation (σ), its Z-score may be calculated by subtracting mean from X and dividing by standard deviation

$$Z = \frac{(X - \mu)}{\sigma}$$

- Interpreting Z score
- A z-score
  - Less than 0 represents an element less than the mean
  - Greater than 0 represents an element greater than the mean
  - Equal to 0 represents an element equal to the mean
  - Equal to 1 represents an element, which is 1 standard deviation greater than the mean
  - Equal to 2 signifies 2 standard deviations greater than the mean; etc.
  - Equal to -1 represents an element, which is 1 standard deviation less than the mean
  - Equal to -2 signifies 2 standard deviations less than the mean; etc.
- If the number of elements in the set is large
  - About 68% of the elements have a z-score between -1 and 1
  - About 95% have a z-score between -2 and 2
  - About 99% have a z-score between -3 and 3

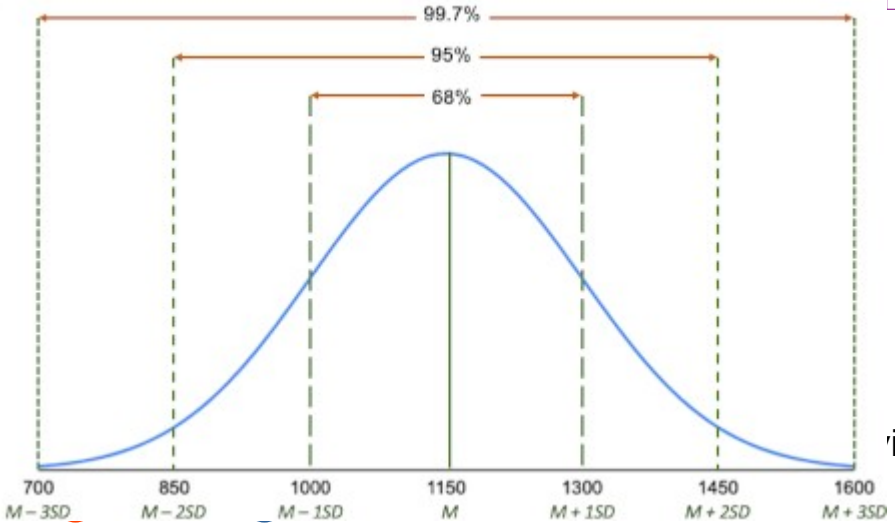
Uses of the Standard Normal Curve

- To find the probability of observations in a distribution falling above or below a given value
- To find the probability that a sample mean significantly differs from a known population mean
- To compare scores on different distributions with different means and standard deviations

Finding Probability Using the Z-distribution

- Each z-score is associated with a probability, or p-value, that tells you the likelihood of values below that z-score occurring
- Example
  - Given the distribution below for SAT scores, find the probability of SAT scores in your sample exceeding 1380

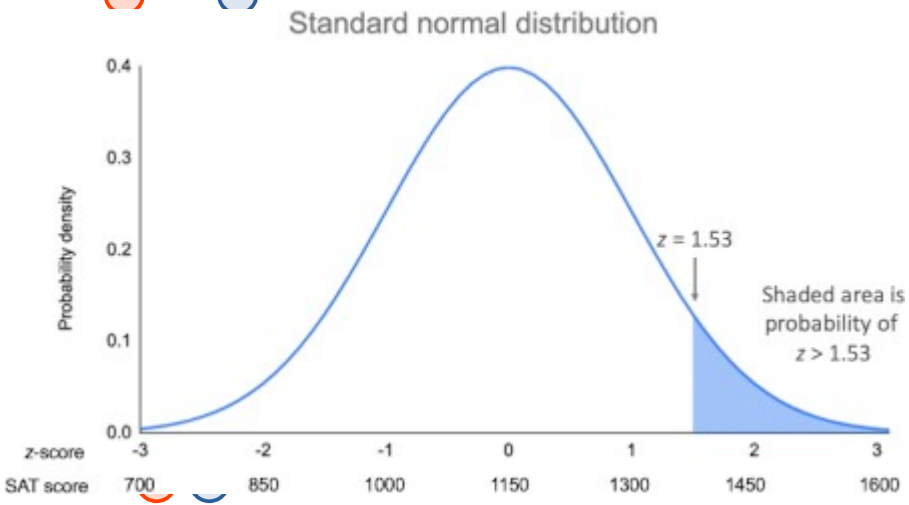
- Mean score (M) = 1150
- Standard deviation (SD) = 150



$$z = \frac{x - \mu}{\sigma} \qquad z = \frac{1\,380 - 1\,150}{150}$$

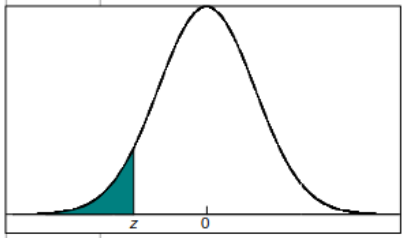
$$z = 1.53$$

Finding Probability Using the Z-distribution



- For a z-score of 1.53, the p-value is 0.937 (statistical tables are available for this)
- This is the probability of SAT scores being 1380 or less (93.7%), and it's the area under the curve left of the shaded area
- The shaded area represents X (score > 1380)
- Probability of x > 1380 = 1 – 0.937 = 0.063
- i.e., it is likely that only 6.3% of SAT scores in your sample exceed 1380

Cumulative Probabilities for the Standard Normal (*Z*) Distribution

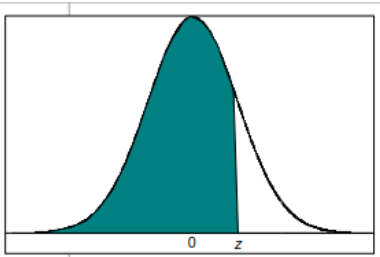


Values in the table correspond to the area under the curve of a standard normal random variable for a value at or below *z*.

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
-5.0	0.0000003									
-4.5	0.000003									
-4.0	0.00003									
-3.5	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002
-3.4	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002
-3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
-3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
-3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
-3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
-2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
-2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
-2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
-2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
-2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
-2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
-2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
-2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
-2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
-2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
-1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
-1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
-1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
-1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
-1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
-1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
-1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
-1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
-1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
-1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
-0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
-0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
-0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
-0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
-0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
-0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
-0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
-0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
-0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
-0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641

IB - IS

Cumulative Probabilities for the Standard Normal (*Z*) Distribution



Values in the table correspond to the area under the curve of a standard normal random variable for a value at or below *z*.

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998
3.5	0.9998									
4.0	0.99997									
4.5	0.999997									
5.0	0.9999997									

Outline

- Data Collection
- Statistics – Review of Some Fundamental Concepts
- Inferential Statistics
- Hypothesis Testing

Descriptive vs Inferential Statistics

- Descriptive Statistics
  - Used to summarize and graph the data for a group (e.g., sample) that you choose
    - Allows you to understand that specific set of observations
  - You simply take a group that you're interested in, record data about the group members, and then use summary statistics and graphs to present the group properties
    - This procedure allows us to gain more insights and visualize the data than simply pouring through row upon row of raw numbers!
  - There is no uncertainty as you are describing only the people or items that you actually measure
    - You're not trying to infer properties about a larger population

Descriptive vs Inferential Statistics

- Common tools of descriptive statistics
  - Central tendency
    - This measure tells you where most values fall
    - Mean and mode
  - Dispersion
    - How far out from the centre do the data extend?
    - Common measures: Range, standard deviation
    - A low dispersion indicates that the values cluster more tightly around the centre
    - Higher dispersion signifies that data points fall further away from the centre
    - Can also graph the frequency distribution
- Skewness
  - Tells you whether the distribution of values is symmetric or skewed
- 5-number summary
  - Highest value in the dataset.
  - Third quartile (Q3)—greater than 75% of the values in the dataset
  - Median or second quartile (Q2)—splits the dataset in half
  - First quartile (Q1)—greater than 25% of the values.
  - Lowest value in the dataset
- Assessing the relationships of paired data using correlation and scatter plots

Descriptive vs Inferential Statistics

- Inferential Statistics
  - Takes data from a sample and makes inferences about the larger population from which the sample was drawn
  - Goal is to draw conclusions from a sample and generalize them to a population
  - We need to have confidence that our sample accurately reflects the population, hence we must
    - Define the population we are studying
    - Draw a representative sample from that population
    - Use analyses that incorporate sampling error
  - Random sampling allows us to have confidence that the sample represents the population
    - Produces statistics, such as the mean, that do not tend to be too high or too low
    - Using a random sample, we can generalize from the sample to the broader population

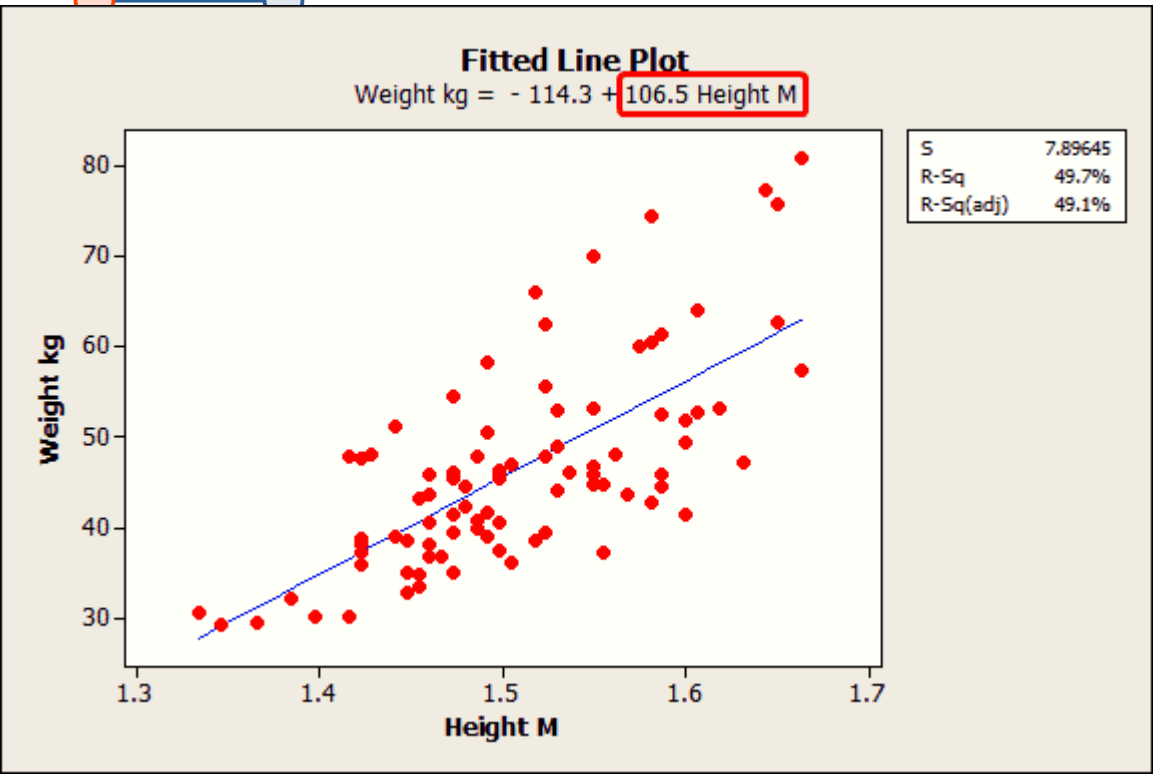
Descriptive vs Inferential Statistics

- Standard analysis tools of inferential statistics
  - The most common methodologies in inferential statistics are
    - Hypothesis tests
    - Confidence intervals
    - Regression analysis
  - These inferential methods can produce similar summary values as descriptive statistics
    - e.g., mean and standard deviation
- Hypothesis tests
  - Use sample data to answer questions like
    - Is the population mean greater than or less than a particular value?
    - Are the means of two or more populations different from each other?

Descriptive vs Inferential Statistics

- Confidence Interval (CI)
  - A range (interval) of values that is likely to contain the value of an unknown population parameter
  - Derived from sample statistics and are calculated using a specified confidence level
  - Useful because population parameters are typically unknown
    - It is usually impossible to measure entire populations
  - By using a sample, you can estimate these parameters
    - Note: The estimates rarely equal the parameter precisely because of random sampling error
    - Inferential statistics procedures can evaluate a sample and incorporate the uncertainty inherent when using samples
      - Inferential statistics use a random sample to draw conclusions about the population
    - Confidence intervals place a margin of error around the point estimate to help us understand how wrong the estimate might be

Descriptive vs Inferential Statistics



- Regression analysis
  - Describes the relationship between a set of independent variables and a dependent variable
  - This analysis incorporates hypothesis tests that help determine whether the relationships observed in the sample data actually exist in the population
  - Example
    - The fitted line plot on the left displays the relationship in the regression model between height and weight in adolescent girls

Outline

- Data Collection
- Statistics – Review of Some Fundamental Concepts
- Inferential Statistics
- Hypothesis Testing

Hypothesis Testing

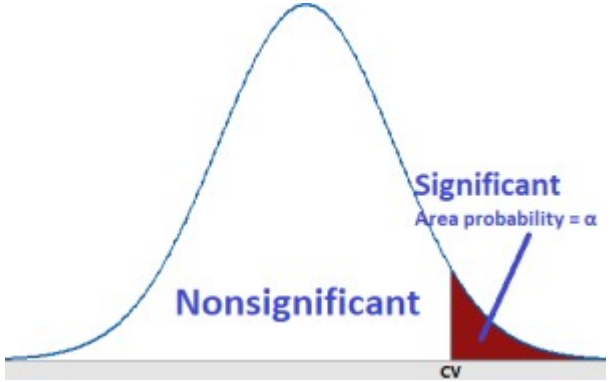
- Critical Values
- Selecting a Statistical Test
- Independent Samples t-Test
- Analysis of Variance
- Potential Errors in the Statistical Decision-Making Process

Critical Values

- A critical value defines regions in the sampling distribution of a test statistic
- Plays a role in both hypothesis tests and confidence intervals
  - In hypothesis tests, critical values determine whether the results are statistically significant
  - For confidence intervals, they help calculate the upper and lower limits
- In both cases, critical values account for uncertainty in sample data you're using to make inferences about a population
- Critical values answer the following questions
  - How different does the sample estimate need to be from the null hypothesis to be statistically significant?
    - A sample estimate is a statistic that estimates a population parameter
  - What is the margin of error around the sample estimate of the population parameter?

Using a Critical Value to Determine Statistical Significance

- Critical values (CV) are the boundary between nonsignificant and significant results in a hypothesis test
- Test statistics that exceed a critical value have a low probability of occurring if the null hypothesis is true
- Therefore, when test statistics exceed these cut-offs, you can reject the null and conclude that the effect exists in the population
  - In other words, they define the rejection regions for the null hypothesis



Significance Level (α)

- The significance level, also known as alpha or α, is determined by the researcher before conducting the experiment
  - It is a measure of the strength of the evidence that must be present in your sample before you will reject the null hypothesis and conclude that the effect is statistically significant
  - It is the probability of rejecting the null hypothesis when it is true
    - e.g., a significance level of 0.05 indicates a 5% risk of concluding that a difference exists when there is no actual difference
    - Lower significance levels indicate that you require stronger evidence before you will reject the null hypothesis

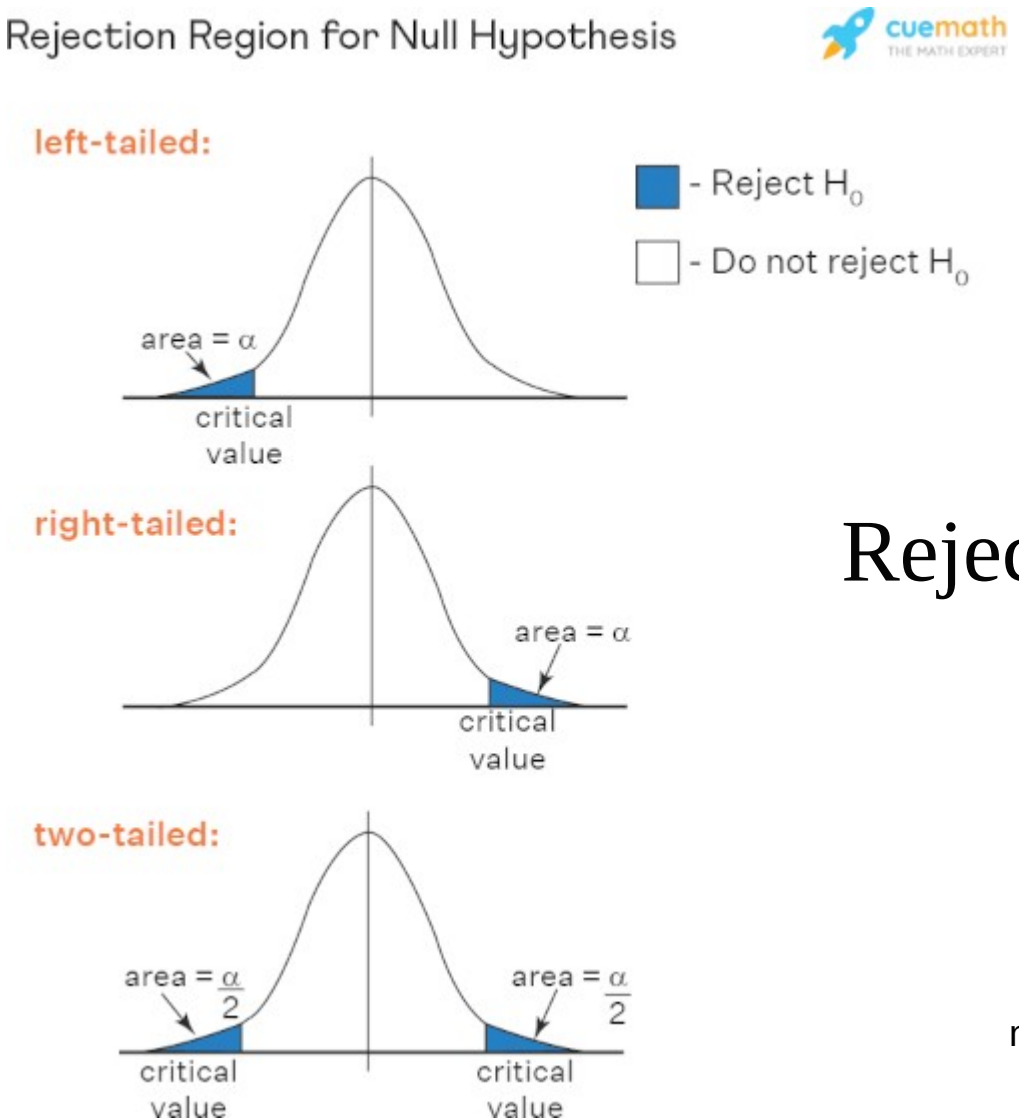
Type I and Type II errors

- In hypothesis tests, two classes of error are possible, Type I and Type II errors
- Type I error: Supporting the alternate hypothesis when the null hypothesis is true
- Type II error: Not supporting the alternate hypothesis when the alternate hypothesis is true
- Courtroom Example
  - Null hypothesis: A man is innocent
  - Alternate hypothesis: He is guilty
  - Type I error: You convict an innocent man, i.e., you support the alternate hypothesis that he is guilty
  - Type II error would be letting a guilty man go free
- So an alpha level is the probability of a type I error (i.e., you reject the null hypothesis when it is true)
- A related term, beta, is the opposite; the probability of rejecting the alternate hypothesis when it is true

Rejection Region

- Also called a critical region
  - An area of a graph where you would reject the null hypothesis (and accept the alternative hypothesis) if your test results fall into that area
- Example: You think that a new fertilizer makes plants grow 50% faster. In order to prove your theory is true, you collect data and do hypothesis testing
- Two Tailed vs One Tailed Tests
  - Which type of test is determined by your scientific hypothesis statement
  - For example, if your statement asks “Is the average growth rate greater than 10cm a day?” that’s a one tailed (right-tailed) test, because you are only interested in one direction (greater than 10cm a day)
  - You could also have a single rejection region for “less than”. For example, “Is the growth rate less than 10cm a day?” (left-tailed test)
  - A two tailed test, with two regions, would be used when you want to know if there’s a difference in both directions (greater than and less than).

Rejection Regions



Rejection Regions and Alpha Levels

- You, as a researcher, choose the alpha level you are willing to accept
- Example, you want to be 95% confident that your results are significant
  - Choose a 5% alpha level (100% – 95%).
    - That 5% level is the rejection region
  - For a one tailed test, the 5% would be in one tail
  - For a two tailed test, the rejection region would be in two tails.
- There are two ways you can test a hypothesis: with a p-value and with a critical value

Rejection Regions and Alpha Levels

- P-value method
  - When you run a hypothesis test (for example, a z test), the result of that test will be a p value (a “probability value.”)
  - This value tells you if your hypothesis statement is probably true or not
  - If the value falls in the rejection region, it means you have statistically significant results; you can reject the null hypothesis
  - If the p-value falls outside the rejection region, it means your results aren’t enough to throw out the null hypothesis
  - In the example of the plant fertilizer, a statistically significant result would be one that shows the fertilizer does indeed make plants grow faster (compared to other fertilizers).
- Critical value method
- Same steps as in p-value method
- However, instead of calculating a p-value you calculate a critical value
- If the value falls inside the region, you reject the null hypothesis.

P-value

- This is the probability that you would obtain the effect observed in your sample, or larger, if the null hypothesis is true for the populations
- P-values are calculated based on your sample data and under the assumption that the null hypothesis is true
- Lower p-values indicate greater evidence against the null hypothesis

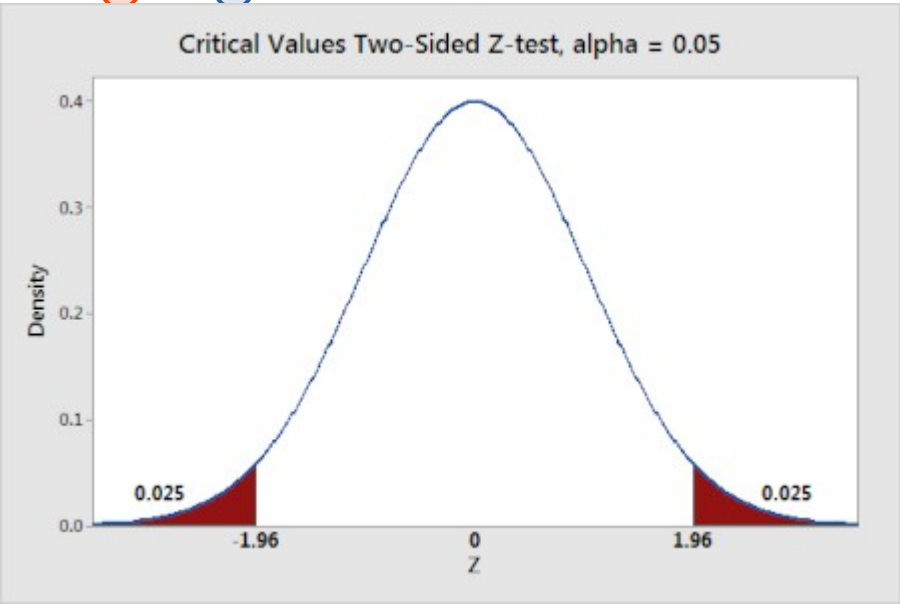
Using Significance Level (α) and p-values

- Use α and p-values during hypothesis testing to help you determine which hypothesis the data support
- Compare your p-value to α
  - If the p-value is less than α, you can reject the null hypothesis and conclude that the effect is statistically significant
  - In other words, the evidence in your sample is strong enough to be able to reject the null hypothesis at the population level

Critical Values

- Critical values depend on your significance level and whether you're performing a one- or two-sided hypothesis
- For illustration in the examples that follow, we'll use a significance level of 0.05
- This value defines how **improbable** the test statistic must be to be significant

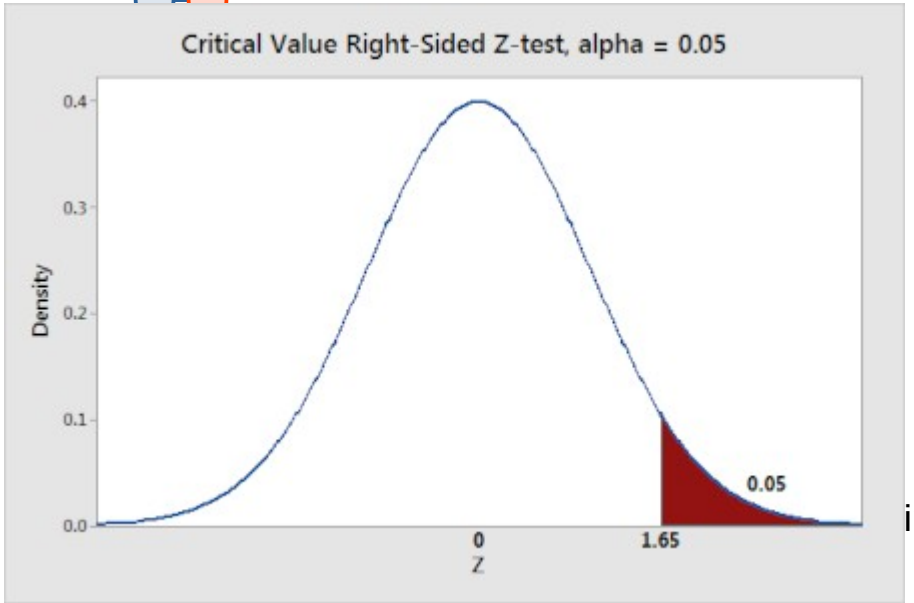
Two-Sided Tests

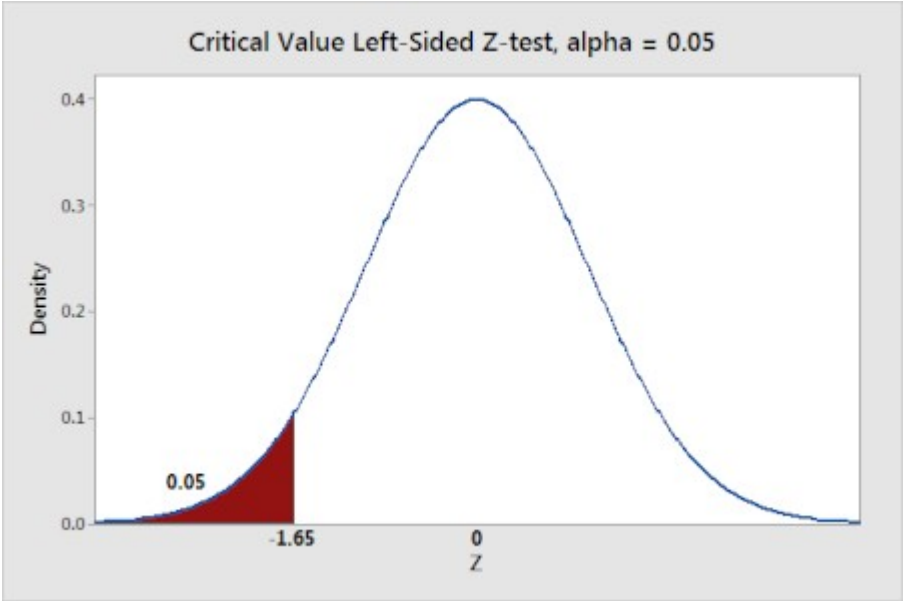


- Two-sided hypothesis tests have two rejection regions
- Consequently, two critical values that define them are needed
  - We must split our significance level in half. Each rejection region has a probability of α/2 (0.05/2 in our example), making the total likelihood for both areas equal the significance level.
- The probability plot on the left displays the critical values and the rejection regions for a two-sided z-test with a significance level of 0.05
  - When the z-score is ≤ -1.96 or ≥ 1.96, it exceeds the cutoff, and your results are statistically significant.

One-Sided Tests

- One-tailed tests have one rejection region and, hence, only one critical value
  - The total α probability goes into that one side
  - The probability plots below display these values for right- and left-sided z-tests
  - These tests can detect effects in only one direction

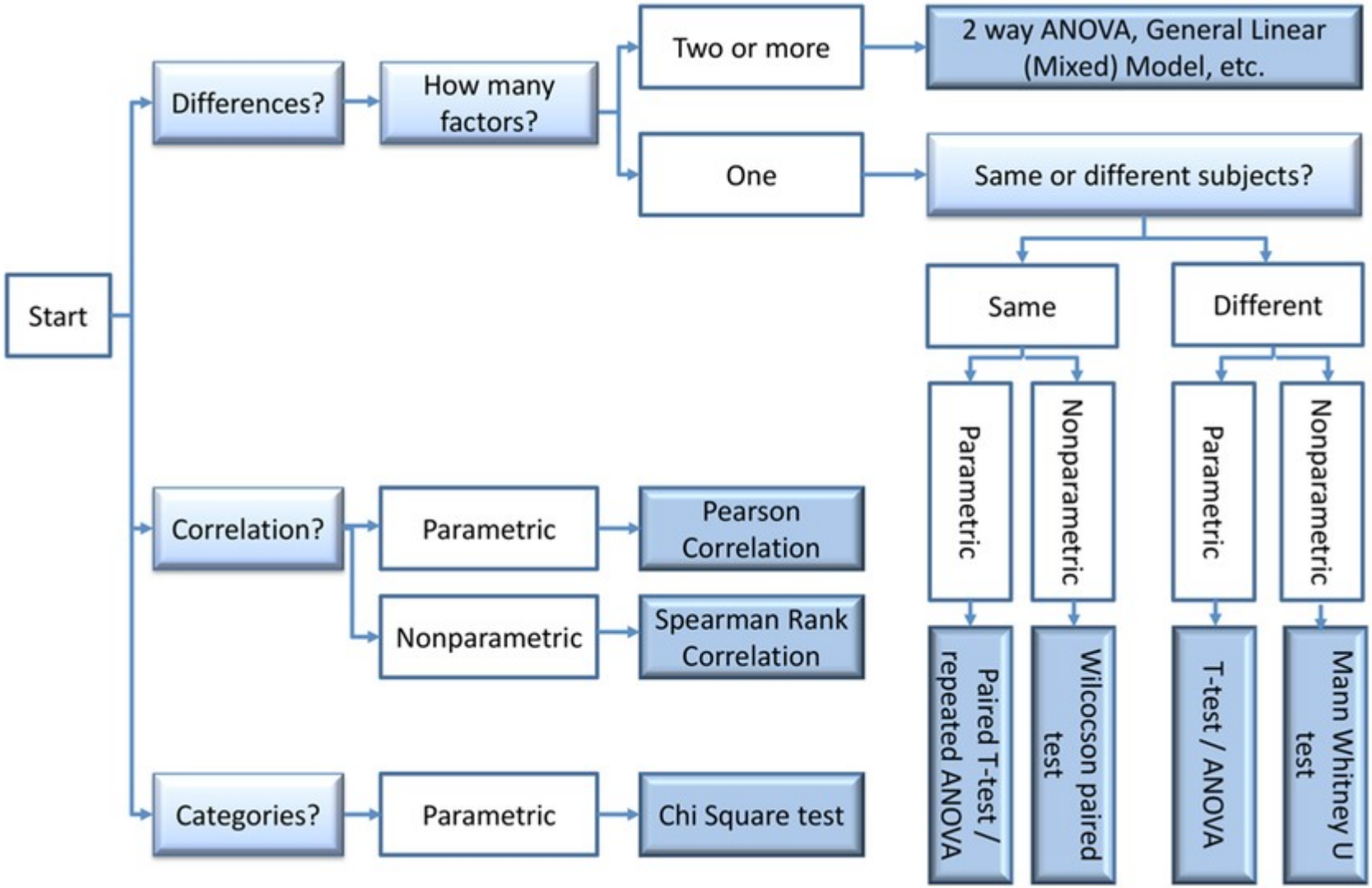




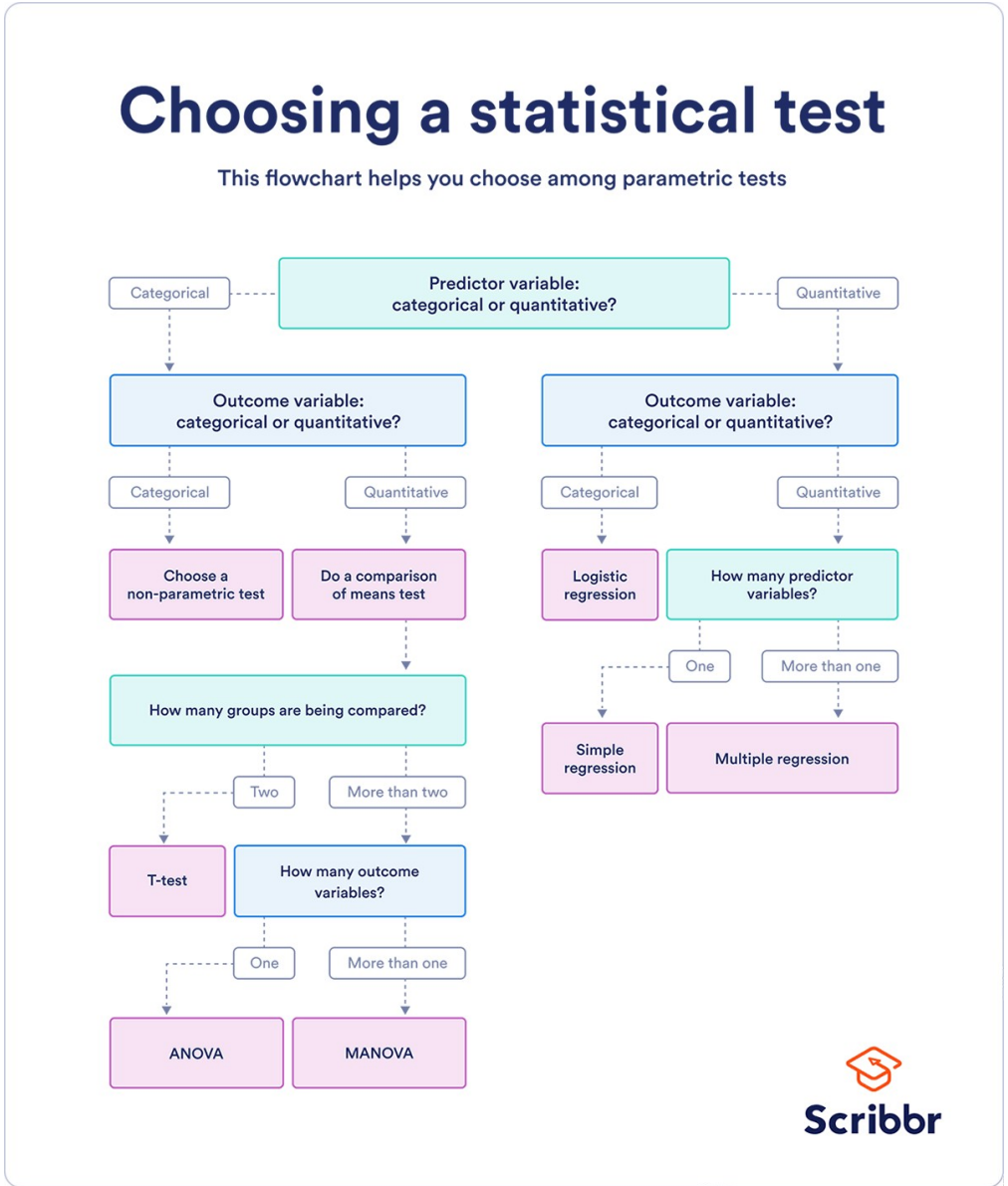
Selecting a Statistical Test

- In the previous chapter, we saw different experimental designs
- After collecting data
  - The data must be analyzed statistically to determine whether difference in the group mean scores is so large that it cannot reasonably be attributed to chance
  - Which statistical test is used to analyze your data is dictated by your experimental design
    - Reason why you must consider the statistical analysis of your data while you design the study
    - The design you use (and the subsequent type of analysis) should be one that answers the research question

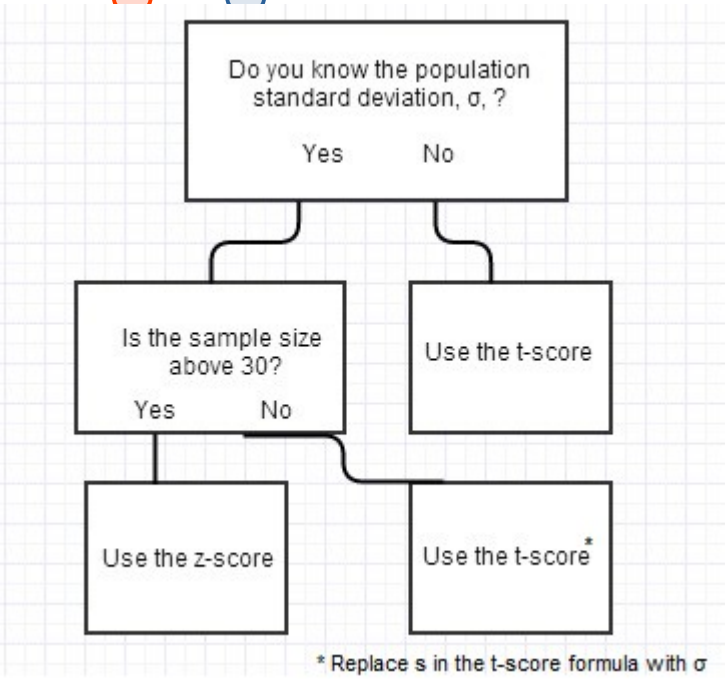
Selecting a Statistical Test



Selecting a Statistical Test



T-Score vs Z-Score



- Before this point, we have been discussing the z-score (and not t-score that the t-test would use). What is the difference?

General rule of thumb for when to use a t-score is when your sample:

- Has a sample size below 30, OR
- Has an unknown population standard deviation

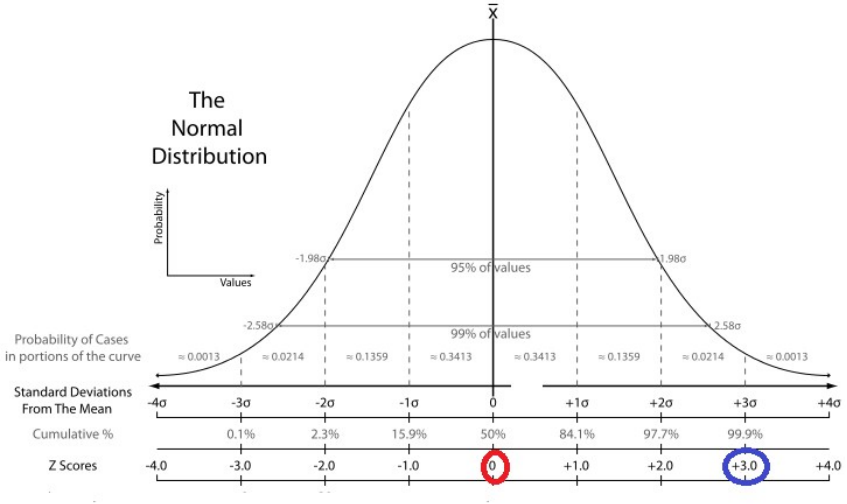
To use the z-score

- You **must** know the standard deviation of the population and
- Your sample size should be above 30

In most cases, we do not know the population standard deviation, and so the t-score (and not the z-score) is used

T-Score vs Z-Score

- As we saw earlier
  - Z-scores are a conversion of individual scores into a standard form
  - The conversion allows you to more easily compare different data
    - It is based on your knowledge about the population's standard deviation and mean
    - A z-score tells you how many standard deviations from the mean your result is
    - You can use your knowledge of normal distributions (like the 68 95 and 99.7 rule) or the z-table to determine what percentage of the population will fall below or above your result
    - Calculated as follows: $z = (X-\mu)/\sigma$
      - σ is the population standard deviation and
      - μ is the population mean
      - Formula doesn't say anything about sample size
      - Rule of thumb: You can use z-score if sample size is above 30



- T-score
  - Like z-scores, t-scores are a conversion of individual scores into a standard form
  - However, t-scores are used when you don't know the population standard deviation; You make an estimate by using your sample
    - $T = (X - \mu) / [\, s/\sqrt{(n)}\,]$ (s is the standard deviation of the sample)
  - For larger sample size (over 30), the t-distribution and z-distribution look pretty much the same. Therefore, you can use either
  - If you do know σ, it doesn't make much sense to use a sample estimate instead of the "real thing", so just substitute σ into the equation in place of s:
    - $T = (X - \mu) / [\, \sigma/\sqrt{(n)}\,]$
  - Use T table, not the Z-table for critical value
  - For sample sizes over 30, you'll get the same result

Hypothesis Testing Example

- We'll consider the study described below left

Research Question: Do environmental conditions predispose an organism (rats) to drug abuse?

Researcher reared rats in either a sterile or a socially enriched environment condition and then assessed the extent to which the rats self-administered two different dosages of cocaine. The amount of self-administration was measured by the number of times the rats pressed a lever that infused a measured amount of cocaine into their jugular veins.

Null hypothesis H_0: Rats do not increase their abuse of cocaine as their access to it increased
Alternative hypothesis H_a: Rats increase their abuse of cocaine as their access to it increased

Method: You train two groups of rats to press a lever that would give either 0.5 mg/kg (Group 1) or 1.0 mg/kg (Group 2) of cocaine

Hypothetical results on the right

mber of Lever Presses Made by Rats Receiving
e of Two Cocaine Dosages

Group 1 (.5 mg/kg dose)	Group 2 (1.0 mg/kg dose)
35	25
25	19
29	22
27	21
24	20
27	21
26	22
25	25
27	29
29	21
26	22
20	20
$\Sigma X_1 = 320$	$\Sigma X_2 = 267$
$\Sigma X_1^2 = 8{,}672$	$\Sigma X_2^2 = 6{,}027$
$\bar{X}_1 = 26.67$	$\bar{X}_2 = 22.25$
$N = 12$	$N = 12$

Testing the Hypothesis

- Which test to use
  - The experiment involves different groups → one independent variable, i.e., factor (dosage of cocaine administered) → different subjects → parametric (i.e., based on assumptions about the distribution of population from which the sample was taken) → T-test/ANOVA
  - Alternatively, predictor variable (categorical, i.e., either 0.5 mg/kg or 1 mg/kg), outcome variable (quantitative, i.e., no of presses), do a comparison of means test, 2 groups → T-test

Independent Samples t-Test

- A statistical test for analyzing the data obtained from two different groups of participants to determine weather the group mean difference score is so large that they could not reasonably be attributed to chance

$$t = \frac{\mu_A - \mu_B}{\sqrt{\left[\frac{\left(\Sigma A^2 - \frac{(\Sigma A)^2}{n_A}\right) + \left(\Sigma B^2 - \frac{(\Sigma B)^2}{n_B}\right)}{n_A + n_B - 2}\right] \cdot \left[\frac{1}{n_A} + \frac{1}{n_B}\right]}}$$

$(\Sigma A)^2$: Sum of data set A, squared
$(\Sigma B)^2$: Sum of data set B, squared
μ_A: Mean of data set A
μ_B: Mean of data set B
ΣA^2: Sum of the squares of data set A
ΣB^2: Sum of the squares of data set B
n_A: Number of items in data set A
n_B: Number of items in data set B

$$t = \frac{26.67 - 22.25}{\sqrt{\frac{8{,}672 - \frac{(320)^2}{12} + 6{,}027 - \frac{(267)^2}{12}}{12 + 12 - 2}\left(\frac{1}{12} + \frac{1}{12}\right)}}$$

$$= \frac{4.42}{1.305}$$

$$= 3.39$$

Independent Samples t-Test

- We have computed a t-statistic (difference of means) of 3.39
- How certain are we that the obtained difference is real?
  - i.e., how certain are we that rats increase their abuse of cocaine as their access to it increased?
    - Notice we need a right-tailed test here
  - Assume a significance level α of 0.05
    - This means that we accept that the difference between the groups is real only if the the difference is so large that it could have occurred by chance 5 times in 100
      - If α of 0.01 was selected, then the difference in means can be expected to occur only 1 time out of 100 by chance
  - To determine if the difference (the t-score) is large enough to be significant, we consult a table of critical values of t, using the appropriate number of degrees of freedom
    - The t-test has N-2 degrees of freedom, where N is the total number of scores, 24 in our example

Critical Regions of t

	P						
one-tail	0.1	0.05	0.025	0.01	0.005	0.001	0.0005
two-tails	0.2	0.1	0.05	0.02	0.01	0.002	0.001
DF							
1	3.078	6.314	12.706	31.821	63.656	318.289	636.578
2	1.886	2.92	4.303	6.965	9.925	22.328	31.6
3	1.638	2.353	3.182	4.541	5.841	10.214	12.924
4	1.533	2.132	2.776	3.747	4.604	7.173	8.61
5	1.476	2.015	2.571	3.365	4.032	5.894	6.869
6	1.44	1.943	2.447	3.143	3.707	5.208	5.959
7	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	1.397	1.86	2.306	2.896	3.355	4.501	5.041
9	1.383	1.833	2.262	2.821	3.25	4.297	4.781
10	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	1.356	1.782	2.179	2.681	3.055	3.93	4.318
13	1.35	1.771	2.16	2.65	3.012	3.852	4.221
14	1.345	1.761	2.145	2.624	2.977	3.787	4.14
15	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	1.337	1.746	2.12	2.583	2.921	3.686	4.015
17	1.333	1.74	2.11	2.567	2.898	3.646	3.965
18	1.33	1.734	2.101	2.552	2.878	3.61	3.922
19	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	1.325	1.725	2.086	2.528	2.845	3.552	3.85
21	1.323	1.721	2.08	2.518	2.831	3.527	3.819
22	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	1.319	1.714	2.069	2.5	2.807	3.485	3.768
24	1.318	1.711	2.064	2.492	2.797	3.467	3.745
25	1.316	1.708	2.06	2.485	2.787	3.45	3.725
26	1.315	1.706	2.056	2.479	2.779	3.435	3.707
27	1.314	1.703	2.052	2.473	2.771	3.421	3.689
28	1.313	1.701	2.048	2.467	2.763	3.408	3.674
29	1.311	1.699	2.045	2.462	2.756	3.396	3.66
30	1.31	1.697	2.042	2.457	2.75	3.385	3.646
60	1.296	1.671	2	2.39	2.66	3.232	3.46
120	1.289	1.658	1.98	2.358	2.617	3.16	3.373
1000	1.282	1.646	1.962	2.33	2.581	3.098	3.3
Inf	1.282	1.645	1.96	2.326	2.576	3.091	3.291

t Table

cum. prob	$t_{.50}$	$t_{.75}$	$t_{.80}$	$t_{.85}$	$t_{.90}$	$t_{.95}$	$t_{.975}$	$t_{.99}$	$t_{.995}$	$t_{.999}$	$t_{.9995}$
one-tail	**0.50**	**0.25**	**0.20**	**0.15**	**0.10**	**0.05**	**0.025**	**0.01**	**0.005**	**0.001**	**0.0005**
two-tails	**1.00**	**0.50**	**0.40**	**0.30**	**0.20**	**0.10**	**0.05**	**0.02**	**0.01**	**0.002**	**0.001**
df											
1	0.000	1.000	1.376	1.963	3.078	6.314	12.71	31.82	63.66	318.31	636.62
2	0.000	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.327	31.599
3	0.000	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.215	12.924
4	0.000	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	0.000	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	0.000	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	0.000	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	0.000	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	0.000	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	0.000	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	0.000	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	0.000	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	0.000	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	0.000	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	0.000	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	0.000	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	0.000	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	0.000	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	0.000	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	0.000	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.552	3.850
21	0.000	0.686	0.859	1.063	1.323	1.721	2.080	2.518	2.831	3.527	3.819
22	0.000	0.686	0.858	1.061	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	0.000	0.685	0.858	1.060	1.319	1.714	2.069	2.500	2.807	3.485	3.768
24	0.000	0.685	0.857	1.059	1.318	1.711	2.064	2.492	2.797	3.467	3.745
25	0.000	0.684	0.856	1.058	1.316	1.708	2.060	2.485	2.787	3.450	3.725
26	0.000	0.684	0.856	1.058	1.315	1.706	2.056	2.479	2.779	3.435	3.707
27	0.000	0.684	0.855	1.057	1.314	1.703	2.052	2.473	2.771	3.421	3.690
28	0.000	0.683	0.855	1.056	1.313	1.701	2.048	2.467	2.763	3.408	3.674
29	0.000	0.683	0.854	1.055	1.311	1.699	2.045	2.462	2.756	3.396	3.659
30	0.000	0.683	0.854	1.055	1.310	1.697	2.042	2.457	2.750	3.385	3.646
40	0.000	0.681	0.851	1.050	1.303	1.684	2.021	2.423	2.704	3.307	3.551
60	0.000	0.679	0.848	1.045	1.296	1.671	2.000	2.390	2.660	3.232	3.460
80	0.000	0.678	0.846	1.043	1.292	1.664	1.990	2.374	2.639	3.195	3.416
100	0.000	0.677	0.845	1.042	1.290	1.660	1.984	2.364	2.626	3.174	3.390
1000	0.000	0.675	0.842	1.037	1.282	1.646	1.962	2.330	2.581	3.098	3.300
z	0.000	0.674	0.842	1.036	1.282	1.645	1.960	2.326	2.576	3.090	3.291
	0%	50%	60%	70%	80%	90%	95%	98%	99%	99.8%	99.9%
	Confidence Level										

Independent Samples t-Test

- From the table of critical values of t
  - 22 degrees of freedom, 1-tailed test, α of 0.05
  - The critical value of t is 1.717
    - If a non-directional hypothesis had been stated, a 2-tailed test of significance would have been use, and the critical value of t for the same significance level would have been 2.074
  - Because the computed value of t (3.39) is larger than the critical value, the likelihood that the difference in the mean score of two the groups occurred by chance is less than 5 in 100
    - Hence, the null hypothesis of no difference between the two groups is rejected
  - But which of the two dose conditions produced the greatest amount of self-administration?
    - Group 1 (the lower dosage), since it has a higher average number of lever presses (26.67 vs 22.25 for Group 2)

Analysis of Variance (ANOVA)

- An extension of the t-test
- Appropriate for analyzing data generated from a research design that uses more than two levels of one independent variable and/or more than one independent variable
- One-Way ANOVA
  - Simplest form of ANOVA
  - Appropriate for data collected on the basis of a simple randomized design

Analysis of Variance (ANOVA)

Number of Lever Presses Made by Rats Receiving One of Three Cocaine Dosages

Group 1 (0.1 mg/kg dose)	Group 2 (0.5 mg/kg dose)	Group 3 (1.0 mg/kg dose)
35	126	67
41	64	40
4	19	32
8	33	10
29	24	45
27	75	22
12	48	41
0	30	20
1	30	24
17	29	40
30	25	16
19	44	15
1	40	21
$\Sigma X_{0.1} = 224$	$\Sigma X_{0.5} = 587$	$\Sigma X_{1.0} = 393$
$\Sigma X^2_{0.1} = 6{,}252$	$\Sigma X^2_{0.5} = 36{,}729$	$\Sigma X^2_{1.0} = 14{,}901$
$\bar{X}_{0.1} = 17.23$	$\bar{X}_{0.5} = 45.15$	$\bar{X}_{1.0} = 30.23$
$N = 13$	$N = 13$	$N = 13$

$\Sigma X_{tot} = 1{,}204$
$\Sigma X_{tot} = 57{,}882$
$\Sigma X_{tot} = 39$

- Assume that based on the results of the 2-group experiment
  - Now hypothesize that the rats will increase their rate of self-administration of the drug as access to cocaine decreases
  - To test this scientific hypothesis, assume you use 3 groups of rats to press a lever to receive
    - 0.1 mg/kg (Group 1)
    - 0.5 mg/kg (Group 2)
    - 1.0 mg/kg (Group 3)

One-Way ANOVA – Important Measures

- We list the important measures, without going through the details
- (1) Total variation in the datasets
  - This is the sum of between-groups variation and within-groups variation
  - (1a) (SS_{bet}) - Between-groups variation
    - Variability arising from the effect of the independent variable and chance factors
  - (1b) (SS_{within}) - Within-groups variation
    - Variability induced by chance factors estimated from the variability of responses of the participants within each experimental group

One-Way ANOVA – Important Measures

- (2) Mean square
  - Variance estimate obtained by dividing a sum of squares by its appropriate degrees of freedom
  - Comprises mean square between (MS_{bet}) and mean square within (MS_{within})
    - (2b) $MS_{bet} = SS_{bet}$/degrees of freedom
    - (2c) $MS_{within} = SS_{within}$/degrees of freedom
- (3) F-ratio
  - The ratio of the two mean square variance estimates
    - F-ratio = MS_{bet}/MS_{within}

ANOVA Summary Table

- A statistical package produces an ANOVA summary table, such as the one below for one-way ANOVA on the current data

Source of Variation	SS	df	MS	F
Between groups	5,076.05	2	2,538.02	5.94
Within groups	15,636.31	36	434.34	
Total	20,712.36	38		

ANOVA Summary Table

- Using the F-ratio, we can determine the actual probability that the group mean differences were due to chance
- Assuming 0.05 significance level, we must determine whether the F-ratio is so large that the results would occur by chance only 5 times in 100
- As with the t-test, we consult a table of critical values (this time F-distribution), using the appropriate degrees of freedom
- From the ANOVA summary table, we have
  - 2 degrees of freedom for the between-groups variation
  - 36 degrees of freedom for the within-groups variation

CRITICAL VALUES for the "F" Distribution, ALPHA = .05.

Denominator DF	Numerator DF 1	2	3	4	5	6	7	8	9	10
1	161.448	199.500	215.707	224.583	230.162	233.986	236.768	238.883	240.543	241.882
2	18.513	19.000	19.164	19.247	19.296	19.330	19.353	19.371	19.385	19.396
3	10.128	9.552	9.277	9.117	9.013	8.941	8.887	8.845	8.812	8.786
4	7.709	6.944	6.591	6.388	6.256	6.163	6.094	6.041	5.999	5.964
5	6.608	5.786	5.409	5.192	5.050	4.950	4.876	4.818	4.772	4.735
6	5.987	5.143	4.757	4.534	4.387	4.284	4.207	4.147	4.099	4.060
7	5.591	4.737	4.347	4.120	3.972	3.866	3.787	3.726	3.677	3.637
8	5.318	4.459	4.066	3.838	3.687	3.581	3.500	3.438	3.388	3.347
9	5.117	4.256	3.863	3.633	3.482	3.374	3.293	3.230	3.179	3.137
10	4.965	4.103	3.708	3.478	3.326	3.217	3.135	3.072	3.020	2.978
11	4.844	3.982	3.587	3.357	3.204	3.095	3.012	2.948	2.896	2.854
12	4.747	3.885	3.490	3.259	3.106	2.996	2.913	2.849	2.796	2.753
13	4.667	3.806	3.411	3.179	3.025	2.915	2.832	2.767	2.714	2.671
14	4.600	3.739	3.344	3.112	2.958	2.848	2.764	2.699	2.646	2.602
15	4.543	3.682	3.287	3.056	2.901	2.790	2.707	2.641	2.588	2.544
16	4.494	3.634	3.239	3.007	2.852	2.741	2.657	2.591	2.538	2.494
17	4.451	3.592	3.197	2.965	2.810	2.699	2.614	2.548	2.494	2.450
18	4.414	3.555	3.160	2.928	2.773	2.661	2.577	2.510	2.456	2.412
19	4.381	3.522	3.127	2.895	2.740	2.628	2.544	2.477	2.423	2.378
20	4.351	3.493	3.098	2.866	2.711	2.599	2.514	2.447	2.393	2.348
21	4.325	3.467	3.072	2.840	2.685	2.573	2.488	2.420	2.366	2.321
22	4.301	3.443	3.049	2.817	2.661	2.549	2.464	2.397	2.342	2.297
23	4.279	3.422	3.028	2.796	2.640	2.528	2.442	2.375	2.320	2.275
24	4.260	3.403	3.009	2.776	2.621	2.508	2.423	2.355	2.300	2.255
25	4.242	3.385	2.991	2.759	2.603	2.490	2.405	2.337	2.282	2.236
26	4.225	3.369	2.975	2.743	2.587	2.474	2.388	2.321	2.265	2.220
27	4.210	3.354	2.960	2.728	2.572	2.459	2.373	2.305	2.250	2.204
28	4.196	3.340	2.947	2.714	2.558	2.445	2.359	2.291	2.236	2.190
29	4.183	3.328	2.934	2.701	2.545	2.432	2.346	2.278	2.223	2.177
30	4.171	3.316	2.922	2.690	2.534	2.421	2.334	2.266	2.211	2.165
31	4.160	3.305	2.911	2.679	2.523	2.409	2.323	2.255	2.199	2.153
32	4.149	3.295	2.901	2.668	2.512	2.399	2.313	2.244	2.189	2.142
33	4.139	3.285	2.892	2.659	2.503	2.389	2.303	2.235	2.179	2.133
34	4.130	3.276	2.883	2.650	2.494	2.380	2.294	2.225	2.170	2.123
35	4.121	3.267	2.874	2.641	2.485	2.372	2.285	2.217	2.161	2.114
36	4.113	3.259	2.866	2.634	2.477	2.364	2.277	2.209	2.153	2.106
37	4.105	3.252	2.859	2.626	2.470	2.356	2.270	2.201	2.145	2.098
38	4.098	3.245	2.852	2.619	2.463	2.349	2.262	2.194	2.138	2.091
39	4.091	3.238	2.845	2.612	2.456	2.342	2.255	2.187	2.131	2.084
40	4.085	3.232	2.839	2.606	2.449	2.336	2.249	2.180	2.124	2.077
41	4.079	3.226	2.833	2.600	2.443	2.330	2.243	2.174	2.118	2.071

- With 2 and 36 degrees of freedom respectively, we find a critical value of 3.259 from the F-distribution table at a significance level of 0.05
- The critical value is less than the computed F-ratio of 5.94
  - Because the F-ratio is more than the critical value
    - We conclude that there is a significant difference between the three groups
    - We reject the null hypothesis and accept the scientific (alternate) hypthesis that the different cocaine dosages significantly affect the number of lever presses made by the rats in the various groups
    - But which group pressed the lever most frequently?
      - We use descriptive statistics (the mean number of lever presses for each group
        - Group 2 (0.5 mg/kg) had the greatest number of responses – mean = 45.15
        - Group 1 (0.1 mg/kg) had the least number of responses – mean = 17.23
        - Group 3 (1.0 mg/kg) had a response rate between Groups 1 and 2 – mean = 30.23

Two-Way ANOVA

- This statistical test is applied to data collected using a factorial design
  - Two or more independent variables studied simultaneously to determine their independent and interactive effects on the dependent variable
- Example
  - An experiment that investigates whether consuming caffeine at an early age sensitizes individuals to cocaine abuse

Two-Way ANOVA

Method
Randomly assign one group of young rats to receive a dose of 1 gm of caffeine per 5kg of body weight in drinking water and another group to receive quinine-laced drinking water each day for a period of 2 months. Both of these substances are bitter, so the taste of the water should be similar in both groups. We end up with one group of rats pre-exposed to caffeine and the other group that is not.

Randomly assign the rats in each of these two groups to receive one of the following doses of cocaine: 0.1 mg/kg, 0.5 mg/kg, or 1.0 mg/kg

	Group 1 (A1) (0.1 mg/kg dose)	Group 2 (A2) (0.5 mg/kg dose)	Group 3 (A3) (1.0 mg/kg dose)	
Caffeine preexposure (B1)	35	64	67	$\Sigma B1 = 763$
	41	19	40	
	14	33	52	$\bar{X}_{B1} = 42.39$
	18	24	60	
	29	75	45	$N_{B1} = 18$
	27	48	72	
Non-exposure to caffeine (B2)	12	30	41	$\Sigma B2 = 433$
	0	30	20	
	1	29	24	$\bar{X}_{B2} = 24.06$
	17	25	40	
	30	44	16	$N_{B2} = 18$
	19	40	15	
	$\Sigma A1B1 = 164$	$\Sigma A2B1 = 263$	$A3B1 = 336$	
	$\Sigma A1B2 = 79$	$\Sigma A2B2 = 198$	$A3B2 = 156$	
	$\Sigma A1 = 243$	$\Sigma A2 = 461$	$\Sigma A3 = 492$	
	$\bar{X}_{A1} = 20.25$	$\bar{X}_{A2} = 38.42$	$\bar{X}_{A3} = 41.00$	
	$N_{A1} = 12$	$N_{A2} = 12$	$N_{A3} = 12$	
	$\Sigma X_{tot} = 1{,}196$	$\Sigma X^2_{tot} = 51{,}884$	$N_{tot} = 36$	

ANOVA Summary Table

- In consulting the 2-way ANOVA summary table, the degrees of freedom of interest are the MS values of the main and interaction effects (2, 1, and 2) on the one hand, and the error MS value (i.e., 30)
- Corresponding critical values are
  - 3.316 for the cocaine main effect
  - 4.171 for the caffeine pretreatment main effect
  - 3.316 for the interaction effect
- Compared to the computed F-ratios
  - Both cocaine and caffeine pretreatment differentially affect lever pressing (i.e., are significant)
  - But the caffeine pretreatment does not differentially influence the effect of cocaine (i.e., there is no interaction between these variables)

ANOVA Summary Table for a Two-Way Analysis of Variance

Source of Variation	SS	df	MS	F
Cocaine	3,068.22	2	1,534.11	8.48
Caffeine pretreatment	3,025.00	1	3,025.00	16.72
Cocaine × caffeine pretreatment	630.00	2	315.00	1.74 ~~1.19~~
Total	12,150.22	30		

- How do cocaine dose and caffeine pretreatment affect self-administration of cocaine?
- Consider the mean scores for each treatment group
  - For the 3 cocaine doses
    - A1 (0.1 mg/kg) had the lowest number of responses – mean = 20.25
    - A3 (1.0 mg/kg) had the highest number of responses – mean = 41
    - A2 (0.5 mg/kg) – mean = 38.42
  - For the caffeine preexposure/non-exposure groups
    - B1 (pretreated with caffeine) had the highest number of responses – mean = 42.39
    - B2 (no caffeine pretreatment) had fewer responses – mean = 24.06

Potential Errors in Statistical the Decision-Making Process